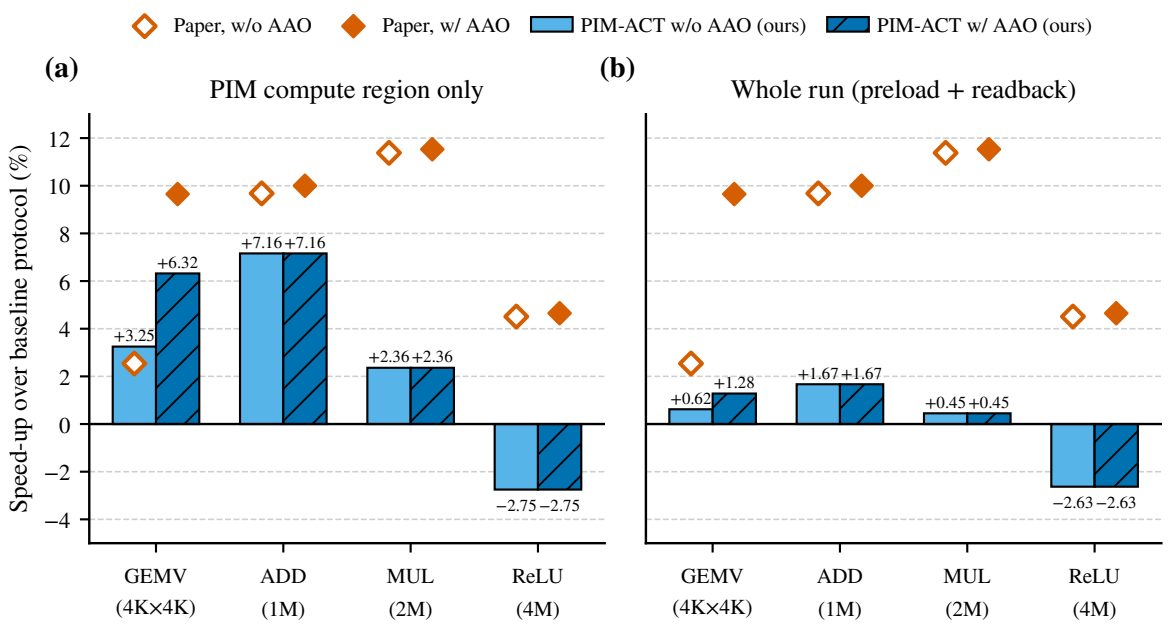
Paper, w/o AAO
Paper, w/ AAO
PIM-ACT w/o AAO (ours)
PIM-ACT w/ AAO (ours)
(a)
PIM compute region only
(b)
Whole run (preload + readback)
Speed-up over baseline protocol (%)
12
10
8
6
4
2
0
−2
−4
+3.25
+6.32
+7.16
+7.16
+2.36
+2.36
−2.75
−2.75
+0.62
+1.28
+1.67
+1.67
+0.45
+0.45
−2.63
−2.63
GEMV (4K×4K)
ADD (1M)
MUL (2M)
ReLU (4M)